SPACEX FALCON 9

First-Stage Landing Prediction

IBM Applied Data Science Capstone

Project Report

Prepared using IBM Skills Network course-prepared SpaceX datasets

Executive Summary

This project evaluates machine-learning classifiers for predicting whether a SpaceX Falcon 9 first stage lands successfully. The analysis used 90 labelled records from the IBM Skills Network course-prepared dataset and 21 encoded predictor columns. The target variable Class represents unsuccessful landing (0) or successful landing (1). Logistic Regression, SVM, Decision Tree and KNN were tuned with three-fold GridSearchCV and evaluated on a held-out 20% test set. The Decision Tree achieved the highest held-out F1-score (0.917) and accuracy (0.889) in this run.

The result is exploratory, not production-grade. The test set contains only 18 observations, and the data is imbalanced. The metrics should be interpreted as results from this specific dataset and split.

1. Introduction

Reusable launch vehicles are designed to recover and reuse major rocket components. Falcon 9 first-stage recovery is an important part of this approach. This project treats landing outcome as a binary classification problem and applies a data-science workflow covering preparation, exploratory analysis, SQL, visual analytics and supervised learning.

2. Dataset and Preparation

The consolidated workflow used IBM Skills Network course-prepared datasets. It did not perform a fresh live SpaceX API extraction or Wikipedia scrape. After excluding records without a target value, 90 labelled rows remained. The target was Class and the feature matrix contained 21 encoded predictor columns.

• Class 0 (unsuccessful): 30 records.

• Class 1 (successful): 60 records.

• Categorical predictors were one-hot encoded; missing numeric values were imputed with the median.

• An 80/20 train-test split was used with random_state = 42.

The class distribution is imbalanced, so accuracy is interpreted together with precision, recall and F1-score.

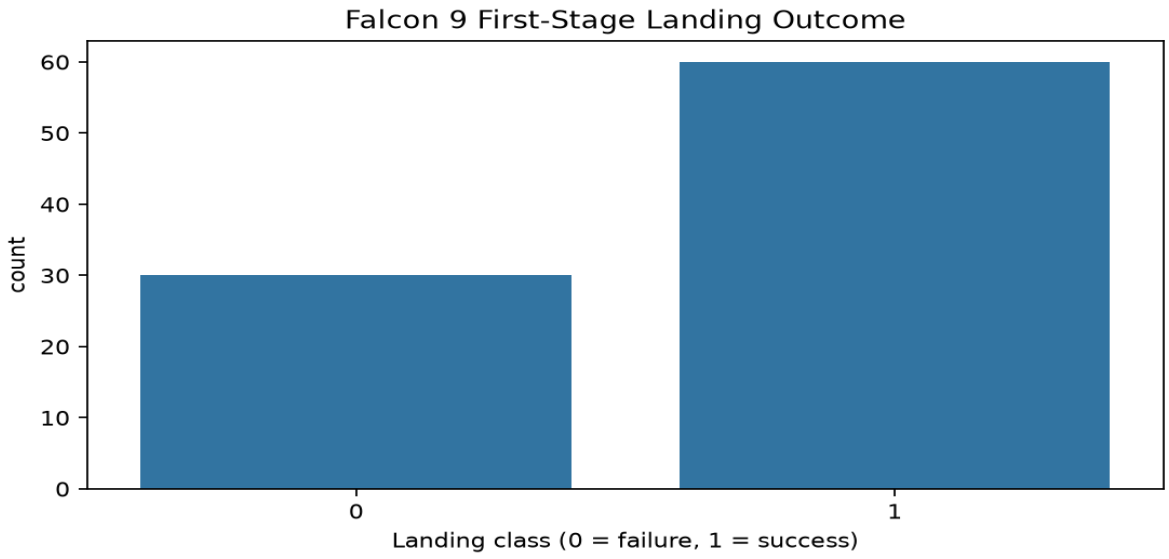


Figure 1. Distribution of labelled landing outcomes.

3. Exploratory Data Analysis

3.1 Payload mass

The payload-mass distributions overlap across the two landing classes. The figure does not indicate that payload mass alone cleanly separates successful from unsuccessful outcomes; several launch characteristics should be considered together.

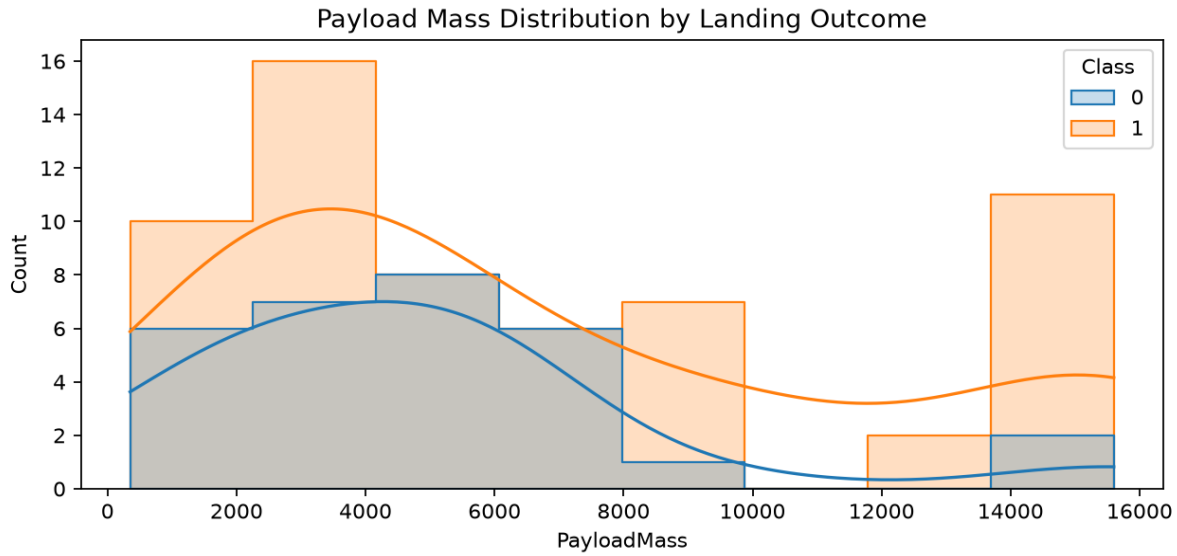


Figure 2. Payload mass distribution by landing class.

3.2 Booster version, launch site and orbit

The categorical plots show counts by outcome. CCAFS SLC 40 has the largest number of launches, followed by KSC LC 39A and VAFB SLC 4E. Orbit groups have uneven sample sizes, so comparisons for rare categories should be treated cautiously.

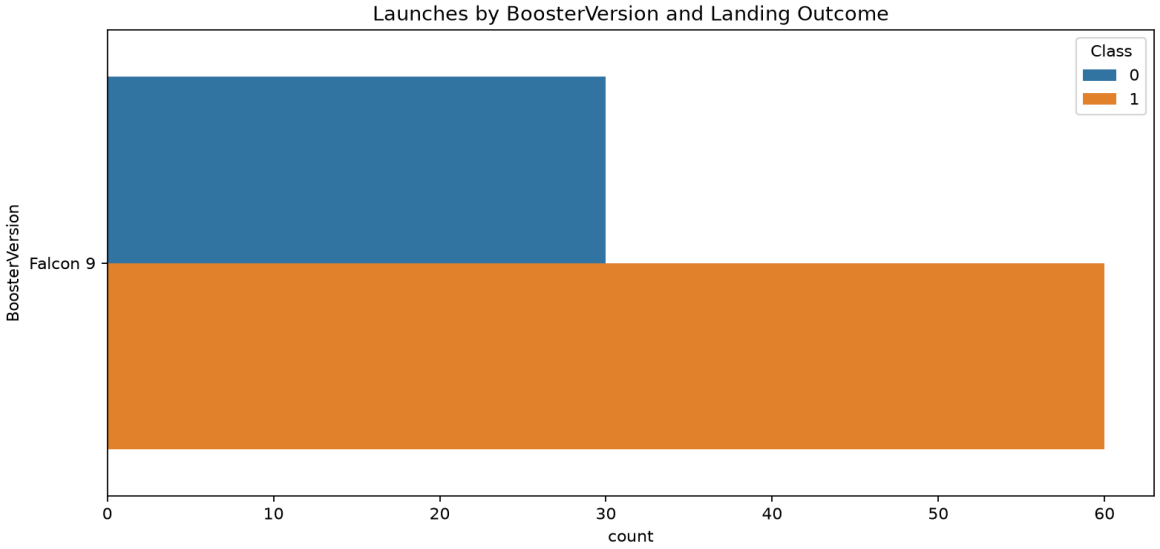


Figure 3. Launches by booster version and landing outcome.

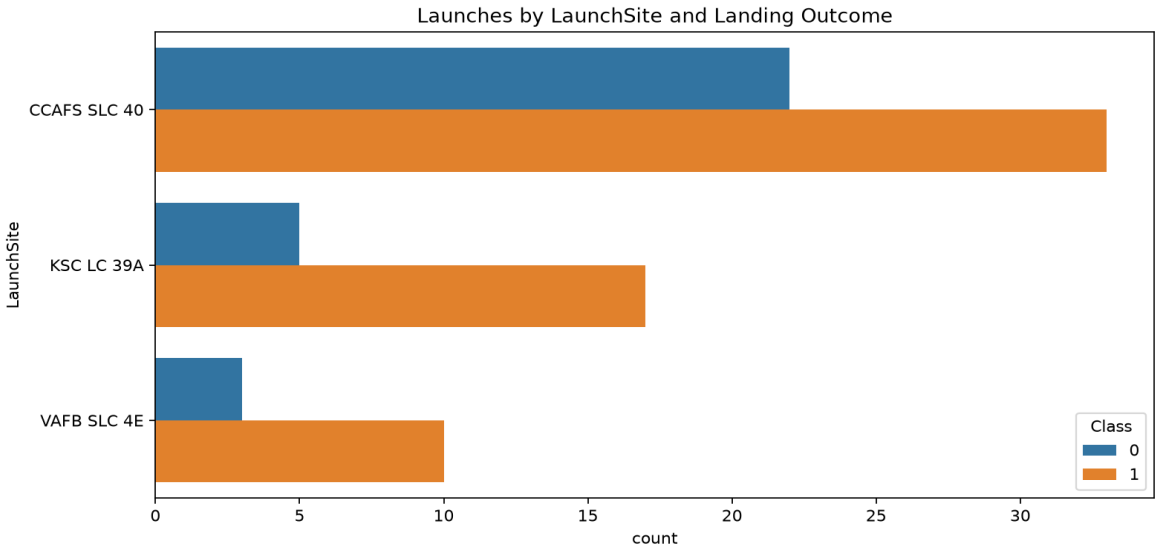


Figure 4. Launches by launch site and landing outcome.

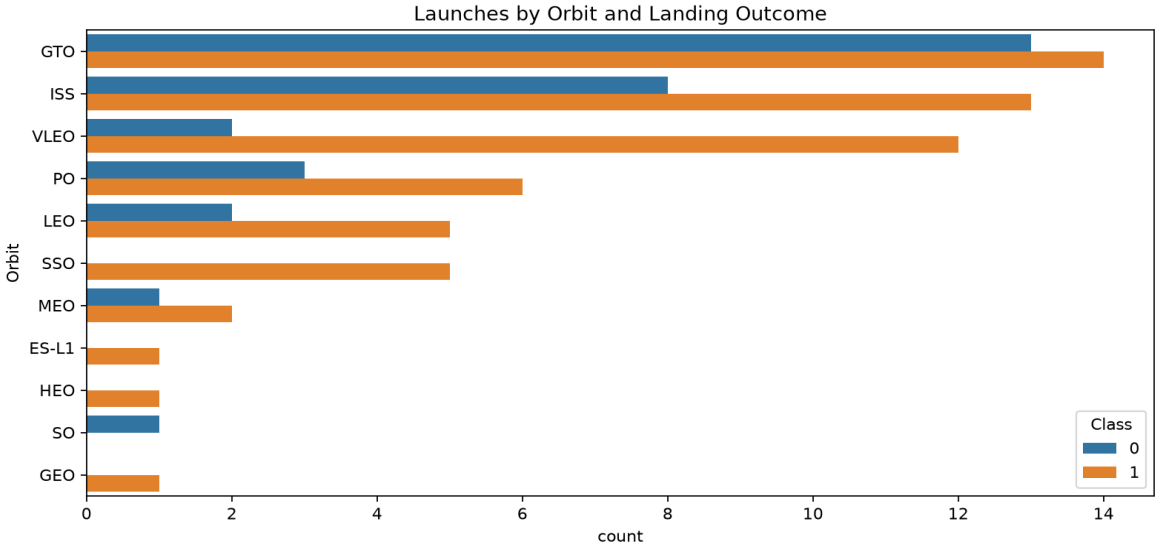


Figure 5. Launches by orbit and landing outcome.

3.3 Correlation analysis

The heatmap summarizes pairwise linear relationships among numeric fields. FlightNumber, Block and ReusedCount show positive associations with some other launch-history variables. Correlation is descriptive and does not establish causation.

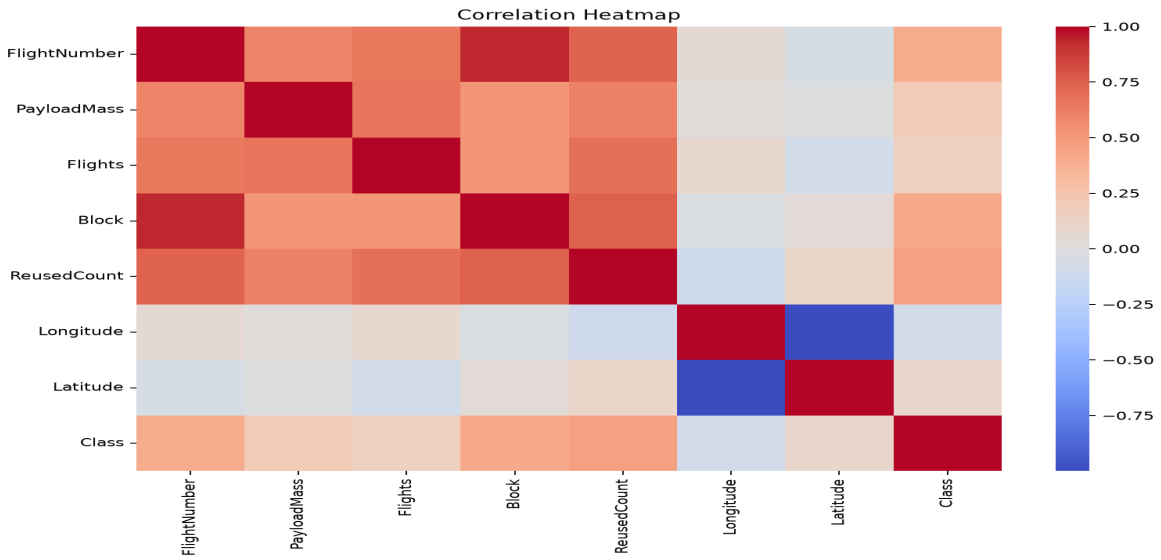


Figure 6. Correlation heatmap.

4. SQL Analysis

The cleaned table was loaded into SQLite. GROUP BY queries counted launches by site and calculated the mean of the binary Class field, which corresponds to the observed Class 1 proportion.

Launch site	Launches	Observed Class 1 rate
KSC LC 39A	22	77.3%
VAFB SLC 4E	13	76.9%
CCAFS SLC 40	55	60.0%

KSC LC 39A has the highest observed rate in this sample (77.3%), closely followed by VAFB SLC 4E (76.9%). CCAFS SLC 40 has the most launches (55) and a 60.0% observed Class 1 rate. These are descriptive sample rates, not causal effects.

For orbit groups, GTO has 27 records and a 51.9% observed Class 1 rate; ISS has 21 records and 61.9%. Rates for groups with one observation are highly uncertain.

5. Interactive Visual Analytics

A Folium map was generated with markers for the three launch sites represented in the data. Popups show the site name, launch count and observed landing rate. The map is saved as outputs/launch_sites_map.html. A Plotly dashboard is saved as outputs/interactive_dashboard.html. External map tiles and browser libraries may require internet access.

6. Machine-Learning Methodology

Four classifiers were evaluated: Logistic Regression, SVM, Decision Tree and KNN. GridSearchCV searched selected hyperparameter combinations using three-fold cross-validation. Each selected estimator was evaluated on the same held-out test set. Scaling was applied within the Logistic Regression, SVM and KNN pipelines.

Model	CV F1	Accuracy	Precision	Recall	Test F1
Decision Tree	0.882	0.889	0.917	0.917	0.917
Logistic Regression	0.864	0.833	0.800	1.000	0.889
KNN	0.889	0.833	0.800	1.000	0.889
SVM	0.889	0.778	0.750	1.000	0.857

The Decision Tree had the highest held-out F1-score (0.917) and test accuracy (88.9%); selected parameters were max_depth = 2 and min_samples_split = 2. Logistic Regression and KNN each scored 0.889 test F1, while SVM scored 0.857. KNN had the highest cross-validation F1 (0.889), showing that CV ranking and held-out ranking differed.

7. Best Model Evaluation

The Decision Tree confusion matrix contained 5 true negatives, 1 false positive, 1 false negative and 11 true positives. It correctly classified 16 of the 18 test examples. The class-wise F1-scores were 0.83 for Class 0 and 0.92 for Class 1.

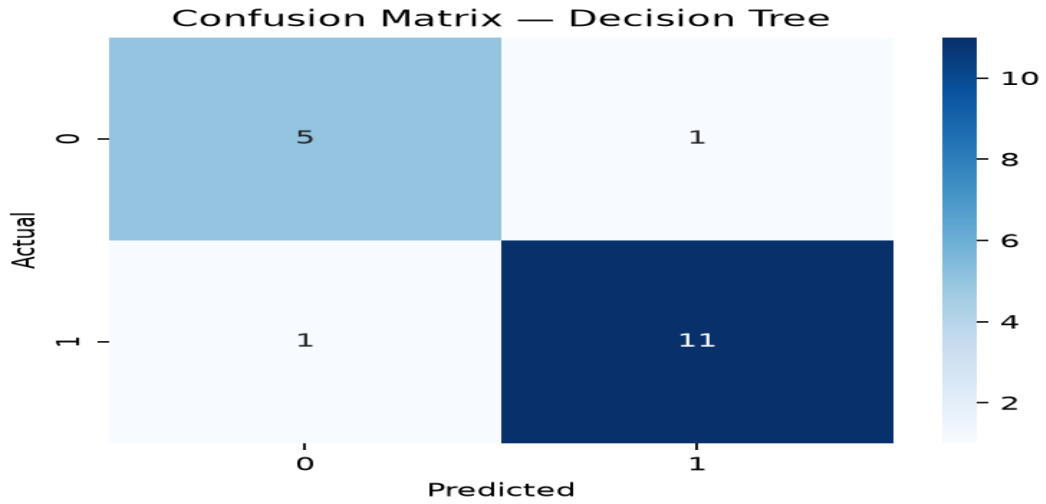


Figure 7. Decision Tree confusion matrix.

There was one error in each direction. Since the test set is small, individual predictions have a noticeable effect on the reported metrics.

8. Conclusion

This project completed exploratory analysis, SQL summaries, interactive visualizations and a comparison of four classification models on the prepared Falcon 9 dataset. The Decision Tree performed best on this run with a held-out F1-score of 0.917 and accuracy of 88.9%. This demonstrates the application of supervised learning to the binary landing-outcome task, but the sample size limits the strength of the conclusion.

9. Limitations and Future Work

• Only 90 labelled records were used, including 18 test observations.

• A single random split can produce unstable performance estimates on a small dataset.

• The course-prepared dataset may not represent all current or future launches.

• Potential feature leakage should be reviewed, particularly for fields that may only be known after launch.

• Future work should add more labelled data, use repeated or nested cross-validation, and test on a chronological holdout set.

• Additional error analysis and probability calibration would help assess prediction reliability.

10. Reproducibility

The consolidated workflow is notebooks/00_RUN_EVERYTHING.ipynb, which calls run_all.py. Dependencies are listed in requirements.txt. Generated CSV files, plots, SQLite database, Folium map and Plotly dashboard are stored in the outputs directory.

Dataset Source

IBM Skills Network — Applied Data Science Capstone course datasets:

https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBM-DS0321EN-SkillsNetwork/datasets/dataset_part_1.csv

https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBM-DS0321EN-SkillsNetwork/datasets/dataset_part_2.csv

https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBM-DS0321EN-SkillsNetwork/datasets/dataset_part_3.csv